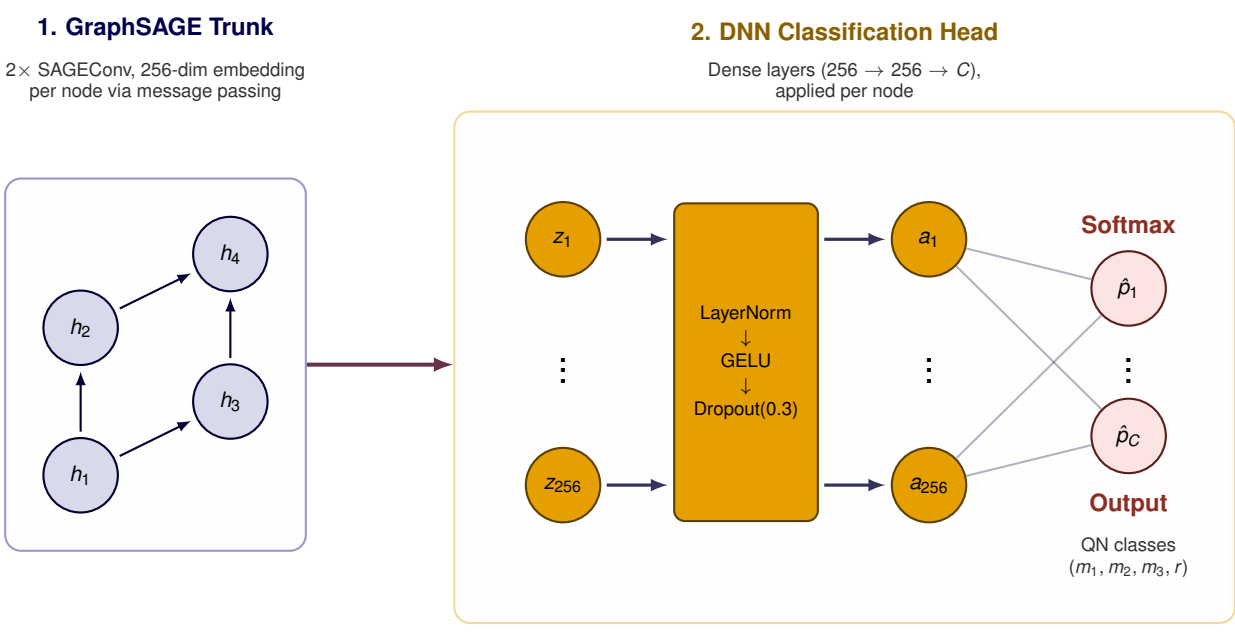
1. GraphSAGE Trunk
2× SAGEConv, 256-dim embedding
per node via message passing
h_4
h_2
h_3
h_1
2. DNN Classification Head
Dense layers (256 → 256 → C),
applied per node
z_1
z_{256}
LayerNorm
↓
GELU
↓
Dropout(0.3)
a_1
a_{256}
Softmax
$\hat{p}_1$
$\hat{p}_C$
Output
QN classes
(m_1, m_2, m_3, r)